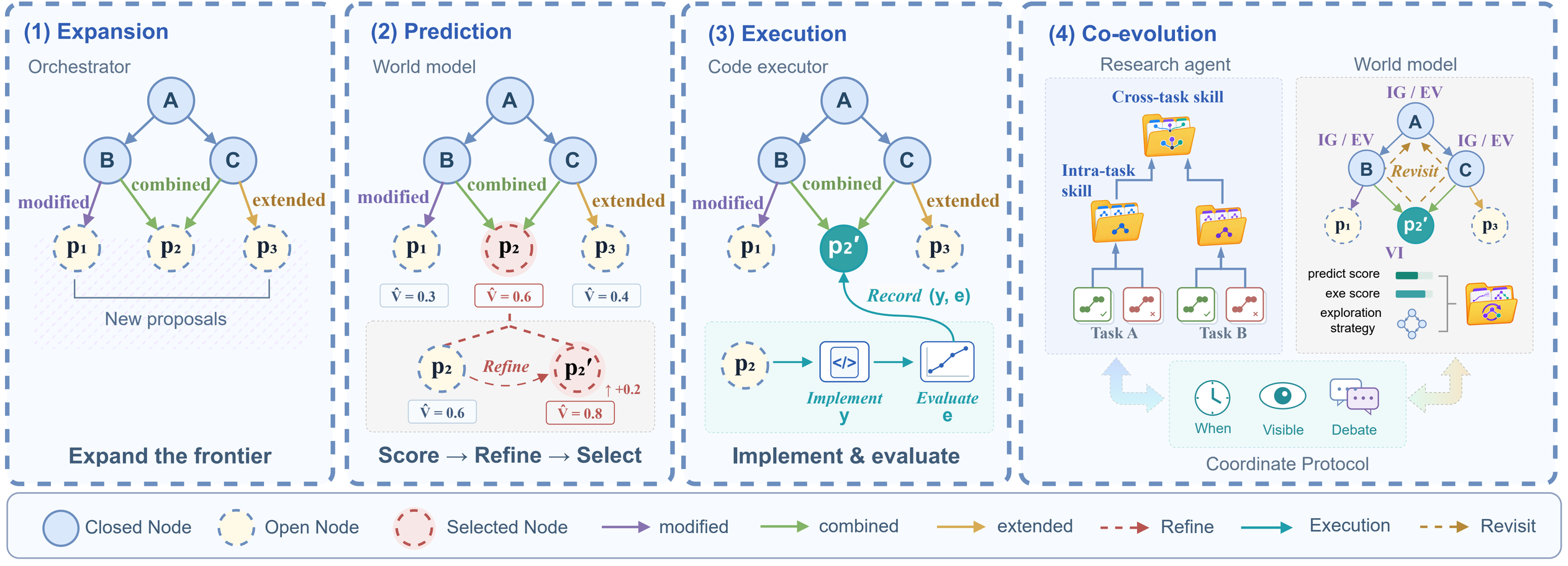

(1) Expansion
Orchestrator
A
B
C
modified
combined
extended
p1
p2
p3
New proposals
Expand the frontier
(2) Prediction
World model
A
B
C
modified
combined
extended
p1
p2
p3
V̂ = 0.3
V̂ = 0.6
V̂ = 0.4
p2
Refine
p2′
↑ +0.2
V̂ = 0.6
V̂ = 0.8
Score → Refine → Select
(3) Execution
Code executor
A
B
C
modified
combined
extended
p1
p2′
p3
Record (y, e)
p2
</>
Implement
y
Evaluate
e
Implement & evaluate
(4) Co-evolution
Research agent
Cross-task skill
Intra-task skill
Task A
Task B
World model
IG / EV
A
IG / EV
IG / EV
B
Revisit
C
p1
p2′
p3
VI
predict score
exe score
exploration strategy
When
Visible
Debate
Coordinate Protocol
Closed Node
Open Node
Selected Node
modified
combined
extended
Refine
Execution
Revisit